EXPERIMENT - 7

AIM

To develop a Servlet-based Online Bus Pass Application System that stores passenger preferences such as theme and language using cookies and retrieves them to provide a personalized user experience.

ALGORITHM

1. Input:

- *The user selects preferences such as theme (light/dark) and language from the bus pass web form.*

2. Steps for Implementing Cookies

Step 1: Create the Preference Form (HTML)

- *Design a webpage for the Online Bus Pass system.*
- *Provide options to select:*
  - *Theme (Light/Dark)*
  - *Language*
- *Add a submit button to send data to the servlet.*

Step 2: Create the Preference Servlet

- *Develop a servlet class to handle form submission.*
- *Retrieve selected values using request.getParameter().*

Step 3: Create Cookies

- *Create cookies for user preferences:*
  - *Theme cookie*
  - *Language cookie*
- *Use: Cookie cookie = new Cookie("name", "value").*

Step 4: Add Cookies to Response

- *Add cookies using response.addCookie(cookie).*
- *Set expiry time using cookie.setMaxAge() for persistence.*

Step 5: Retrieve Cookies on Future Visits

- *Use request.getCookies() to get stored cookies.*
- *Loop through cookies and identify theme and language values.*

Step 6: Display Preferences in Bus Pass System

- *Show selected theme and language on the webpage.*
- *Apply theme styling and display content accordingly.*

Step 7: Handle Default Preferences

- *If cookies are not available:*
  - *Set default theme and language*
  - *Display standard bus pass interface*

CODE:

<!DOCTYPE html>

```
<html>
<head>
<title>EXP 7 - Bus Pass Preferences</title>

<style>
body{
    margin:0;
    font-family: Arial;
    background: linear-gradient(135deg,#1e3c72,#2a5298);
    color:white;
}

header{
    background:#1e88e5;
    padding:15px;
    text-align:center;
    font-size:24px;
    font-weight:bold;
}

.container{
    width:400px;
    margin:80px auto;
    padding:25px;
    background: rgba(255,255,255,0.15);
    border-radius:12px;
    box-shadow:0 0 15px rgba(0,0,0,0.3);
```

```
    text-align:center;
}

label{
    display:block;
    margin-top:10px;
    margin-bottom:5px;
}

select{
    width:100%;
    padding:8px;
    border-radius:5px;
    border:none;
}

button{
    width:100%;
    padding:10px;
    margin-top:15px;
    border:none;
    border-radius:25px;
    background: linear-gradient(45deg,#00e5ff,#00c853);
    color:white;
    cursor:pointer;
}
</style>
```

```
</head>

<body>

<header>
   Online Bus Pass Application System
</header>

<div class="container">

<h2>EXP 7 - Set Preferences</h2>

<form action="EXP7Servlet" method="post">

<label>Select Theme:</label>
<select name="theme">
   <option value="light">Light</option>
   <option value="dark">Dark</option>
</select>

<label>Select Language:</label>
<select name="language">
   <option value="English">English</option>
   <option value="Tamil">Tamil</option>
</select>
```

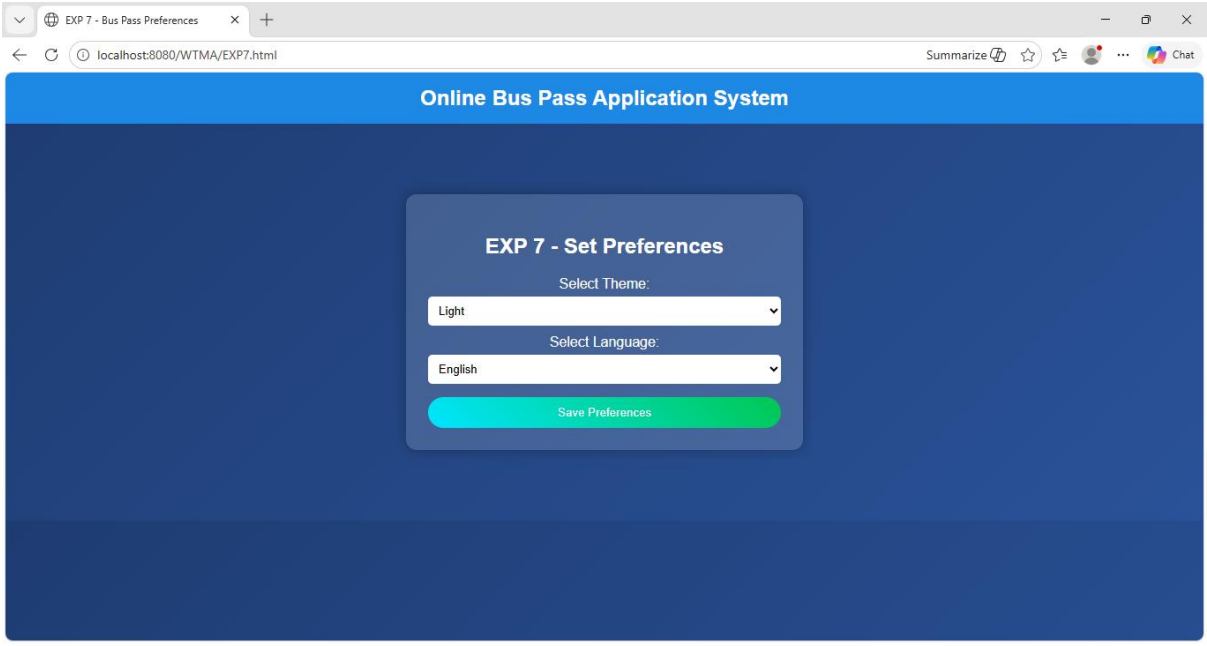
```
<button type="submit">Save Preferences</button>

</form>

</div>

</body>
</html>
package WTMA;

import jakarta.servlet.*;
import jakarta.servlet.http.*;
import jakarta.servlet.annotation.*;
import java.io.*;

@WebServlet("/EXP7Servlet")
public class EXP7Servlet extends HttpServlet {

   protected void doPost(HttpServletRequest request,
HttpServletResponse response)
          throws ServletException, IOException {

       String theme = request.getParameter("theme");
       String language = request.getParameter("language");

       Cookie themeCookie = new Cookie("theme", theme);
       Cookie langCookie = new Cookie("language", language);
```

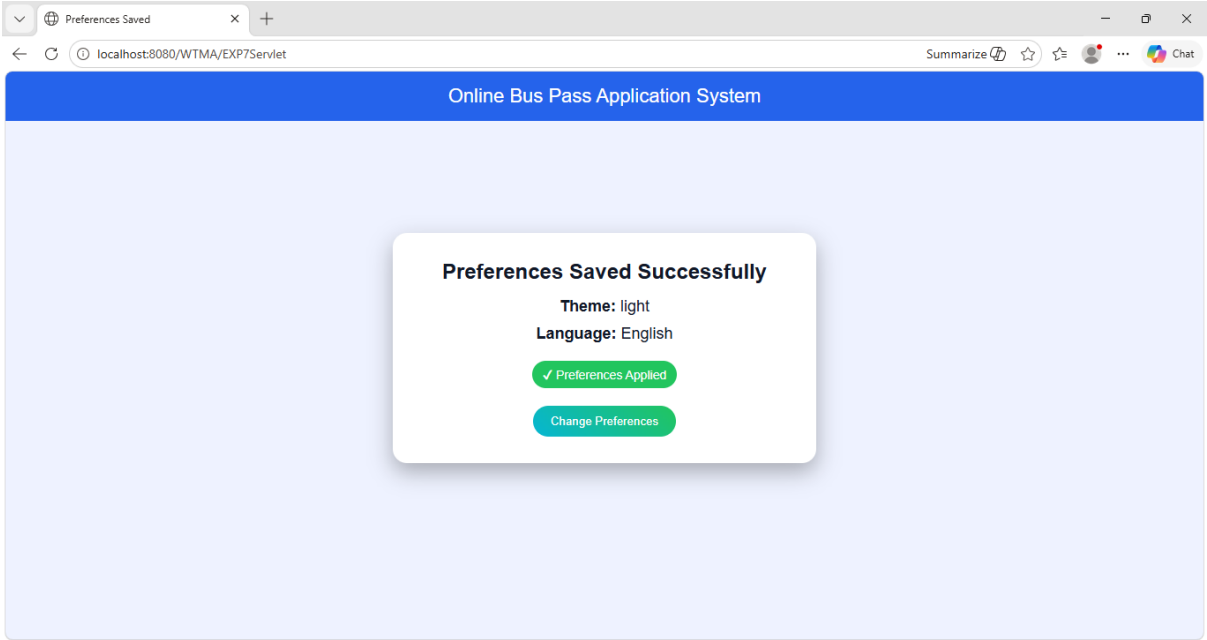
```
        themeCookie.setMaxAge(60*60*24);
        langCookie.setMaxAge(60*60*24);

        response.addCookie(themeCookie);
        response.addCookie(langCookie);

        response.sendRedirect("EXP7Servlet");
    }

    protected void doGet(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter pw = response.getWriter();

        String theme = "light";
        String language = "English";

        Cookie[] cookies = request.getCookies();

        if(cookies != null){
            for(Cookie c : cookies){
                if(c.getName().equals("theme")) theme = c.getValue();
                if(c.getName().equals("language")) language =
c.getValue();
            }
        }
```

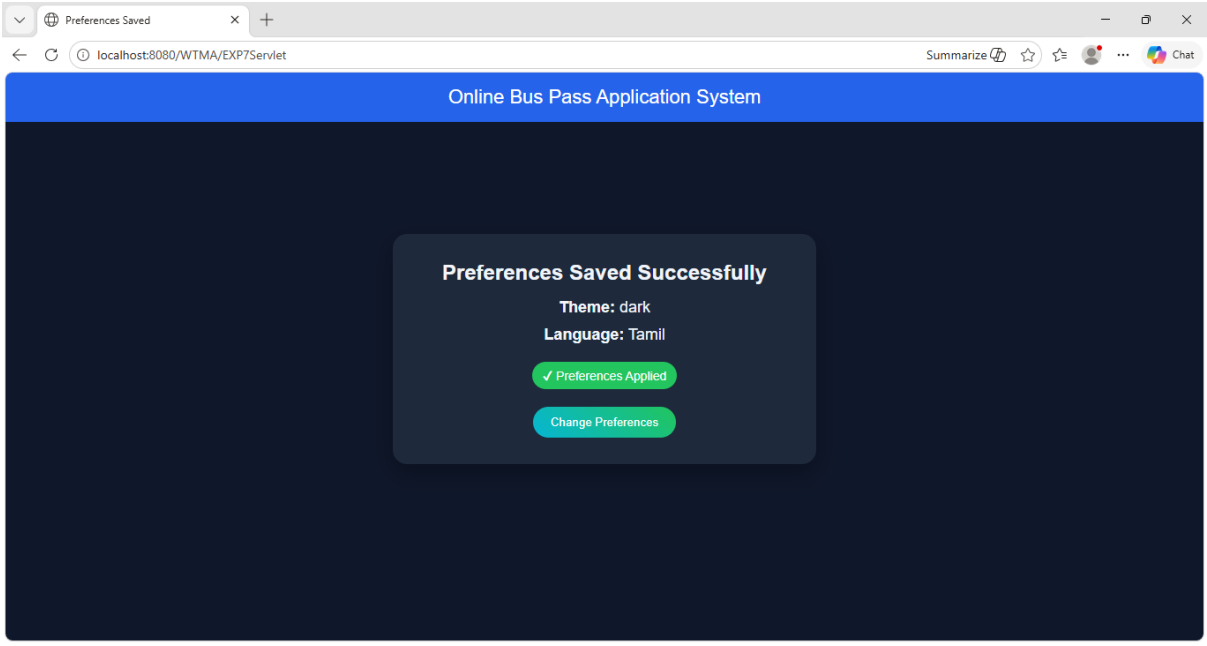
```
String bg = theme.equals("dark") ? "#0f172a" : "#eef2ff";
String card = theme.equals("dark") ? "#1e293b" : "white";
String text = theme.equals("dark") ? "#f1f5f9" : "#111827";

pw.println("<html><head><title>Preferences Saved</title>");

pw.println("<style>");
pw.println("body{margin:0;font-family:Arial;background:"+bg+";color:"+text+";}");
pw.println("header{background:#2563eb;padding:15px;text-align:center;color:white;font-size:22px;}");
pw.println(".container{display:flex;justify-content:center;align-items:center;height:80vh;}");
pw.println(".card{width:420px;padding:30px;border-radius:16px;background:"+card+";box-shadow:0 10px 25px rgba(0,0,0,0.3);text-align:center;}");
pw.println(".title{font-size:24px;margin-bottom:15px;font-weight:bold;}");
pw.println(".info{margin:10px 0;font-size:18px;}");
pw.println(".badge{display:inline-block;padding:6px 12px;border-radius:20px;background:#22c55e;color:white;font-size:14px;margin-top:10px;}");
pw.println(".btn{margin-top:20px;padding:10px 20px;border:none;border-radius:25px;background:linear-gradient(45deg,#06b6d4,#22c55e);color:white;cursor:pointer;}");
pw.println(".btn:hover{transform:scale(1.05);}");
pw.println("</style>");

pw.println("</head><body>");
```

```
        pw.println("<header>Online Bus Pass Application
System</header>");


        pw.println("<div class='container'>");
        pw.println("<div class='card'>");


        pw.println("<div class='title'>Preferences Saved
Successfully</div>");


        pw.println("<div class='info'><b>Theme:</b> " + theme +
"</div>");
        pw.println("<div class='info'><b>Language:</b> " + language +
"</div>");


        pw.println("<div class='badge'>✓Preferences Applied</div>");


        pw.println("<br><button class='btn'
onclick=\"window.location.href='EXP7.html'\">Change
Preferences</button>");


        pw.println("</div>");
        pw.println("</div>");


        pw.println("</body></html>");
    }
}
```

OUTPUT:


EXP 7 - Bus Pass Preferences
localhost:8080/WTMA/EXP7.html
Summarize
Chat
Online Bus Pass Application System
EXP 7 - Set Preferences
Select Theme:
Light
Select Language:
English
Save Preferences


Preferences Saved
localhost:8080/WTMA/EXP7Servlet
Summarize
Chat
Online Bus Pass Application System
Preferences Saved Successfully
Theme: light
Language: English
✓ Preferences Applied
Change Preferences



RESULT:

The Servlet program was successfully implemented to store user preferences using cookies. The application was able to retrieve and display the stored preferences on subsequent visits, providing a personalized experience to the user. Cookies were effectively used to maintain state information across multiple requests.